

CSE 420 - Compiler Design Assignment 01 (Section 18)

Set B Problems

Problem 1: $R_{21} = (d \mid e)^* b(c \mid \epsilon)(da \mid ec)(a \mid c)^*$

Problem 2: $R_{22} = (a \mid c)^* d(e \mid \epsilon)(ae \mid cd)(a \mid d)^*$

Problem 3: $R_{23} = (d \mid c)^* a(b \mid \epsilon)(db \mid ca)(d \mid a)^*$

Problem 4: $R_{24} = (d \mid e)^* c(b \mid \epsilon)(db \mid ec)(d \mid c)^*$

Problem 5: $R_{25} = (e \mid d)^* a(b \mid \epsilon)(eb \mid da)(e \mid d)^*$

Problem 6: $R_{26} = (d \mid e)^* b(a \mid \epsilon)(da \mid ec)(d \mid c)^*$

Problem 7: $R_{27} = (b \mid c)^* a(d \mid \epsilon)(bd \mid ac)(b \mid a)^*$

Problem 8: $R_{28} = (d \mid e)^* b(c \mid \epsilon)(dc \mid eb)(d \mid e)^*$

Problem 9: $R_{29} = (a \mid c)^* (de \mid ca)(d \mid \epsilon)c(e \mid a)^*$

Problem 10: $R_{30} = (d \mid c)^* (ab \mid cd)(a \mid \epsilon)d(b \mid c)^*$

Problem 11: $R_{31} = (d \mid e)^* (cb \mid ed)(c \mid \epsilon)e(b \mid d)^*$

Problem 12: $R_{32} = (e \mid d)^* (ab \mid de)(a \mid \epsilon)d(b \mid e)^*$

Problem 13: $R_{33} = (d \mid e)^* (ba \mid ed)(b \mid \epsilon)e(a \mid d)^*$

Problem 14: $R_{34} = (b \mid c)^* (da \mid cb)(d \mid \epsilon)c(a \mid b)^*$

Problem 15: $R_{35} = (d \mid e)^* (bc \mid ed)(b \mid \epsilon)e(c \mid d)^*$

Problem 16: $R_{36} = (a \mid c)^* d(ac \mid ed)(c \mid \epsilon)(d \mid e)^*$

Problem 17: $R_{37} = (d \mid c)^* a(dc \mid ba)(c \mid \epsilon)(a \mid b)^*$

Problem 18: $R_{38} = (d \mid e)^* c(de \mid bc)(e \mid \epsilon)(c \mid b)^*$

Problem 19: $R_{39} = (e \mid d)^* a(ed \mid ab)(d \mid \epsilon)(a \mid b)^*$

Problem 20: $R_{40} = (d \mid e)^* b(de \mid ab)(e \mid \epsilon)(b \mid a)^*$

Self Study

Study Chapter 3, Sections **3.9.1–3.9.5** (Direct method to construct a DFA from a regular expression using: syntax tree, *nullable*, *firstpos*, *lastpos*, *followpos*, and Algorithm 3.36 / Fig. 3.62).

Submission Format

- **Handwritten submission only.** (No PDF required.)

General Instructions

- Alphabet: $\Sigma = \{a, b, c, d, e\}$ and ϵ denotes the empty string.

- Use the **augmented** regular expression $(R)\#$ (append endmarker $\#$) as in Sec. 3.9.1.
- Follow the textbook method: **syntax tree** $\rightarrow$ compute **nullable/firstpos/lastpos** (Sec. 3.9.3) $\rightarrow$ compute **followpos** (Sec. 3.9.4) $\rightarrow$ construct DFA using **Algorithm 3.36** (Sec. 3.9.5 / Fig. 3.62).

Problem Assignment

Let L be the **last two digits** of your student ID (an integer from 00 to 99).

$$r = L \bmod 20$$

- If $r = 0$, you are assigned **Problem 20**.
- Otherwise, you are assigned **Problem r** .

Example 1 (Nonzero remainder)

Student ID: 20231234 $\Rightarrow L = 34$

$34 \bmod 20 = 14 \Rightarrow$ Assigned: **Problem 14**.

Example 2 (Zero remainder)

Student ID: 20234560 $\Rightarrow L = 60$

$60 \bmod 20 = 0 \Rightarrow$ Assigned: **Problem 20**.

What to Submit

Your handwritten solution must include:

1. Syntax tree of $(R)\#$ with positions labeled for each non- ϵ leaf (including $\#$).
2. Tables (or clear annotations) showing *nullable*, *firstpos*, *lastpos* for each node.
3. *followpos* table for each numbered position.
4. DFA construction:
 - Start state = *firstpos*(*root*)
 - States as sets of positions
 - Transitions using Fig. 3.62 (union of *followpos*)
 - Mark accepting states